Assignment 1

Himanshu Muley

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Setup

```
#Load the required packages
library(inspectdf)
library(lubridate)
library(ggplot2)
library(dplyr)
library(forcats)
library(tidyverse)
library(tidyr)
library(tinytex)
library(stringr)
```

Q1. Loading the data

```
# Your student number goes here
ysn = 1895211
# Calculate your student number modulo 3
filenum <- ysn %% 3
filenum
```

```
## [1] 0
```

```
filename <- paste0("./data/archery_",filenum,".csv")
filename
```

```
## [1] "./data/archery_0.csv"
```

```
# Read in the data
tibble_data <- read_csv("D:\\UNI OF
ADELAIDE\\Trimester\\Data_taming\\archery_0.csv", na= c("NA", "XX"))
head(n=10, tibble_data)
```

```
## # A tibble: 10 × 9
##    name    experienced started session_1 session_2 session_3 session_4
session_5
##    <chr>   <chr>       <chr>   <chr>     <chr>     <chr>     <chr>
<chr>
##  1 Jasper  No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  2 Eva     Yes         10-Nov… Target h… Target h… Target h… Target h…
Target h…
```

```
##  3 Oscar   Yes         05-Mar… Target h… Target h… Target h… Target h…
Target h…
##  4 Millie  Yes         17-Feb… Target h… Target h… Target h… Target h…
Target h…
##  5 Vincent No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  6 Hugo    No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  7 Hamish  No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  8 Lennox  Yes         04-Nov… Target h… Target h… Target h… Target h…
Target h…
##  9 Sophie  Yes         10-Oct… Target h… Target h… Target h… Target h…
Target h…
## 10 Theo    Yes         01-Apr… Target h… Target h… Target h… Target h…
Target h…
## # ℹ 1 more variable: session_6 <chr>

# Display the first 10 lines of the data
```

Q2. Taking a random sample of archers

```
#
set.seed(ysn)
sample_archers <-sample_n(tibble_data, 100, replace = FALSE)
head(n=10, sample_archers)
```

```
## # A tibble: 10 × 9
##    name    experienced started session_1 session_2 session_3 session_4
session_5
##    <chr>   <chr>       <chr>   <chr>     <chr>     <chr>     <chr>
<chr>
##  1 Hallie  Yes         10-Oct… Target h… Target h… Target h… Target h…
Target h…
##  2 Luke    No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  3 Elias   Yes         02-Apr… Target h… Target h… Target h… Target h…
Target h…
##  4 Isabel… Yes         15-Dec… Target h… Target h… Target h… Target h…
Target h…
##  5 Elizab… No          01-Feb… Target h… Target h… Target h… Target h…
Target h…
##  6 Summer  Yes         03-Jun… Target h… Target h… Target h… Target h…
Target h…
##  7 George  Yes         18-Nov… Target h… Target h… Target h… Target h…
Target h…
##  8 Charlie Yes         19-Apr… Target h… Target h… Target h… Target h…
Target h…
##  9 Adeline Yes         18-Mar… Target h… Target h… Target h… Target h…
Target h…
```

```
## 10 Eloise  Yes         20-Nov… Target h… Target h… Target h… Target h…
Target h…
## # ℹ 1 more variable: session_6 <chr>
```

Q3. Tidying the data

Q3(a). Converting from wide form to long form

```
#

tidying_the_data <- gather(sample_archers, key = "session", value = "result",
session_1:session_6)
```

Q3(b). Replacing `result` with 2 new columns

```
#
tidying_the_data <-tidying_the_data %>%
mutate(
  shots = str_match(result, "(\\d+)shots")[,2],
  hits = str_match(result, "(\\d+)times")[,2]
 )  %>% relocate("shots", .before = result) %>%
  relocate("hits", .before = result) %>% select(name:hits)
```

Q3(c). Replace data in `session` column with session number

```
#
tidying_the_data <- tidying_the_data %>%
  mutate(session = str_match(session, "session_(\\d+)")[,2])
```

Q3(d). Replace “Yes/No” with “TRUE/FALSE”

```
#
tidying_the_data <- tidying_the_data %>%
  mutate(experienced = fct_recode(tidying_the_data$experienced, "FALSE"="No",
"TRUE"="Yes"))
head(n=10, tidying_the_data)
```

```
## # A tibble: 10 × 6
##    name      experienced started     session shots hits
##    <chr>     <fct>       <chr>       <chr>   <chr> <chr>
##  1 Hallie    TRUE        10-Oct-2022 1       <NA>  <NA>
##  2 Luke      FALSE       01-Feb-2024 1       <NA>  <NA>
##  3 Elias     TRUE        02-Apr-2023 1       <NA>  <NA>
##  4 Isabelle  TRUE        15-Dec-2021 1       <NA>  <NA>
##  5 Elizabeth FALSE       01-Feb-2024 1       <NA>  <NA>
##  6 Summer    TRUE        03-Jun-2022 1       <NA>  <NA>
##  7 George    TRUE        18-Nov-2022 1       <NA>  <NA>
##  8 Charlie   TRUE        19-Apr-2023 1       <NA>  <NA>
##  9 Adeline   TRUE        18-Mar-2022 1       <NA>  <NA>
## 10 Eloise    TRUE        20-Nov-2022 1       <NA>  <NA>
```

Q4. Identifying data types

Name - This is a Categorical nominal variable because it's just a label used to identify someone, it does not show any ranking.

Experienced - this data is Categorical nominal variable as it has only 2 categories True and False where order or ranking does not matter.

Started - This data has dates in it so this is a quantitative continuous variable as time can't be categorized like nominal or ordinal, time keeps going on without any breaks.

Session - This variable type is Categorical ordinal as it can be put in a order or ranking.

Shots - Shots can be considered as quantitative discrete variable as they are unique to every archer which makes it discrete Hits- This can also be considered as quantitative discrete variable because it records all the different hits made by different archers making it a discrete value

Q5. Taming the data

```
#
tidying_the_data$shots <- as.integer(tidying_the_data$shots)
tidying_the_data$hits <- as.integer(tidying_the_data$hits)
tidying_the_data$session <- as.ordered(tidying_the_data$session)
tidying_the_data$started <- dmy(tidying_the_data$started)
head(n=10, tidying_the_data)
```

```
## # A tibble: 10 × 6
##    name      experienced started    session shots  hits
##    <chr>     <fct>       <date>     <ord>   <int> <int>
##  1 Hallie    TRUE        2022-10-10 1          NA    NA
##  2 Luke      FALSE       2024-02-01 1          NA    NA
##  3 Elias     TRUE        2023-04-02 1          NA    NA
##  4 Isabelle  TRUE        2021-12-15 1          NA    NA
##  5 Elizabeth FALSE       2024-02-01 1          NA    NA
##  6 Summer    TRUE        2022-06-03 1          NA    NA
##  7 George    TRUE        2022-11-18 1          NA    NA
##  8 Charlie   TRUE        2023-04-19 1          NA    NA
##  9 Adeline   TRUE        2022-03-18 1          NA    NA
## 10 Eloise    TRUE        2022-11-20 1          NA    NA
```

Q5.b) Confirming the remaining varaiables to the "tame data" conventions

```
#
tidying_the_data <- tidying_the_data %>% mutate(started =
dmy(tidying_the_data$started))
tidying_the_data <-
```

```
tidying_the_data%>%mutate(session=as.ordered(tidying_the_data$session))
head(n=10, tidying_the_data)
```

```
## # A tibble: 10 × 6
##    name      experienced started session shots  hits
##    <chr>     <fct>       <date>  <ord>   <int> <int>
##  1 Hallie    TRUE        NA      1          NA    NA
##  2 Luke      FALSE       NA      1          NA    NA
##  3 Elias     TRUE        NA      1          NA    NA
##  4 Isabelle  TRUE        NA      1          NA    NA
##  5 Elizabeth FALSE       NA      1          NA    NA
##  6 Summer    TRUE        NA      1          NA    NA
##  7 George    TRUE        NA      1          NA    NA
##  8 Charlie   TRUE        NA      1          NA    NA
##  9 Adeline   TRUE        NA      1          NA    NA
## 10 Eloise    TRUE        NA      1          NA    NA
```

Q6. Adding two new columns to the tidy dataset

```
tidying_the_data$days_experience <- dmy('01/02/2024') -
tidying_the_data$started
tidying_the_data$accuracy <- tidying_the_data$hits/tidying_the_data$shots
head(n=10,tidying_the_data)
```

```
## # A tibble: 10 × 8
##    name      experienced started session shots  hits days_experience
accuracy
##    <chr>     <fct>       <date>  <ord>   <int> <int> <drtn>
<dbl>
##  1 Hallie    TRUE        NA      1          NA    NA NA days
NA
##  2 Luke      FALSE       NA      1          NA    NA NA days
NA
##  3 Elias     TRUE        NA      1          NA    NA NA days
NA
##  4 Isabelle  TRUE        NA      1          NA    NA NA days
NA
##  5 Elizabeth FALSE       NA      1          NA    NA NA days
NA
##  6 Summer    TRUE        NA      1          NA    NA NA days
NA
##  7 George    TRUE        NA      1          NA    NA NA days
NA
##  8 Charlie   TRUE        NA      1          NA    NA NA days
NA
##  9 Adeline   TRUE        NA      1          NA    NA NA days
NA
## 10 Eloise    TRUE        NA      1          NA    NA NA days
NA
```

Q7. displaying the sample statistics for the numerical value in the dataset

```
inspect_num(tibble_data)
```

```
## # A tibble: 0 × 10
## # ℹ 10 variables: col_name <chr>, min <dbl>, q1 <dbl>, median <dbl>,
## #   mean <dbl>, q3 <dbl>, max <dbl>, sd <dbl>, pcnt_na <dbl>, hist <list>
```

#Q.8 Data set of expereinced archers

```
data_filtered <- tidying_the_data %>% filter(experienced == TRUE)
head(data_filtered, n = 10)
```

```
## # A tibble: 10 × 8
##    name     experienced started session shots  hits days_experience
accuracy
##    <chr>    <fct>       <date>  <ord>   <int> <int> <drtn>
<dbl>
##  1 Hallie   TRUE        NA      1          NA    NA NA days
NA
##  2 Elias    TRUE        NA      1          NA    NA NA days
NA
##  3 Isabelle TRUE        NA      1          NA    NA NA days
NA
##  4 Summer   TRUE        NA      1          NA    NA NA days
NA
##  5 George   TRUE        NA      1          NA    NA NA days
NA
##  6 Charlie  TRUE        NA      1          NA    NA NA days
NA
##  7 Adeline  TRUE        NA      1          NA    NA NA days
NA
##  8 Eloise   TRUE        NA      1          NA    NA NA days
NA
##  9 Grace    TRUE        NA      1          NA    NA NA days
NA
## 10 Eva      TRUE        NA      1          NA    NA NA days
NA
```

Q9. Graph who can practice shooting the best

```
ggplot(data_filtered, aes(x = days_experience, y = accuracy)) + geom_point()
+ geom_smooth()
```


accuracy
days_experience